

Assignment 1: Student Attendance Manager (Set + List)

Objective

Store student attendance where:

- Duplicate entries are not allowed
- Order of attendance must be preserved

Tasks

1. Accept student roll numbers using a `List`.
2. Convert the list into a `LinkedHashSet`.
3. Display the final attendance list.

Concepts Used

`List`, `Set`, `LinkedHashSet`

Assignment 2: Online Exam Waiting System (Queue)

Objective

Implement a waiting queue for students.

Tasks

1. Add 5 students to a queue.
2. Remove 2 students after they are served.
3. Display the next student in line.

Concepts Used

`Queue`, `LinkedList`

Assignment 3: Email Subscription System (Set)

Objective

Ensure only unique email IDs are stored.

Tasks

1. Accept 10 email IDs.
2. Store them in a `HashSet`.
3. Display total unique subscribers.

Concepts Used

`HashSet`, `Set`

Assignment 4: Shopping Cart Application (List)

Objective

Manage products in a shopping cart.

Tasks

1. Add items to an `ArrayList`.
2. Remove one item.
3. Update quantity of an item.
4. Display the final cart.

Concepts Used

`ArrayList`, `List`

Assignment 5: Hospital Token System (PriorityQueue)

Objective

Treat emergency patients first.

Tasks

1. Add normal and emergency patients.
2. Use `PriorityQueue` to give emergency higher priority.
3. Display treatment order.

Concepts Used

Assignment 6: Remove Duplicates from List

Problem

Given a list of integers, remove duplicates and display them in **sorted order**.

Tasks

1. Store numbers in a `List`.
 2. Convert to `TreeSet`.
 3. Print the result.
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Assignment 7: Queue Operations Simulator

Problem

Simulate queue operations.

Tasks

1. Enqueue 5 numbers.
 2. Dequeue 2 numbers.
 3. Display remaining elements.
 4. Show front element using `peek()`.
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Assignment 8: First Non-Repeated Character Finder

Problem

Find the first non-repeating character in a string.

Tasks

1. Add characters to a `Queue`.
2. Track repeats using a `Set`.
3. Display the first unique character.

Assignment 9: Unique Word Counter

Problem

Count unique words in a sentence.

Tasks

1. Read a sentence from user.
2. Split into words.
3. Store in `HashSet`.
4. Print count of unique words.

Assignment 10: Ticket Counter Simulation

Problem

Simulate people standing in a ticket queue.

Tasks

1. Add customers to a queue.
 2. Serve them one by one.
 3. Display service order.
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